



SC1006

Computer Organisation and Architecture

Communication and User Interface

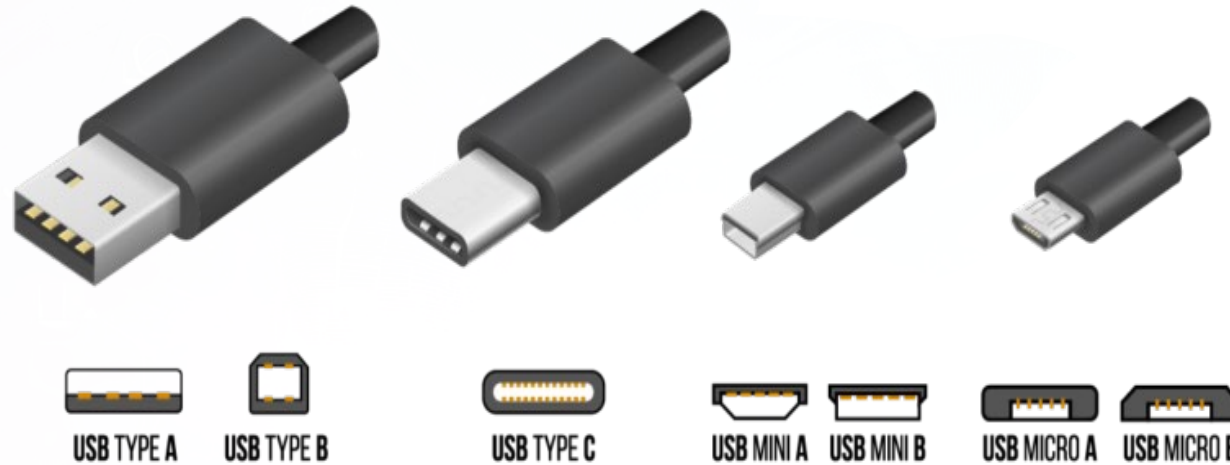
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An abstract digital graphic on the left side of the slide. It features a stylized map of Australia in the background, overlaid with a complex network of glowing blue and white lines and nodes, suggesting a global or digital communication theme. Various geometric shapes like hexagons and circles are scattered throughout the design.

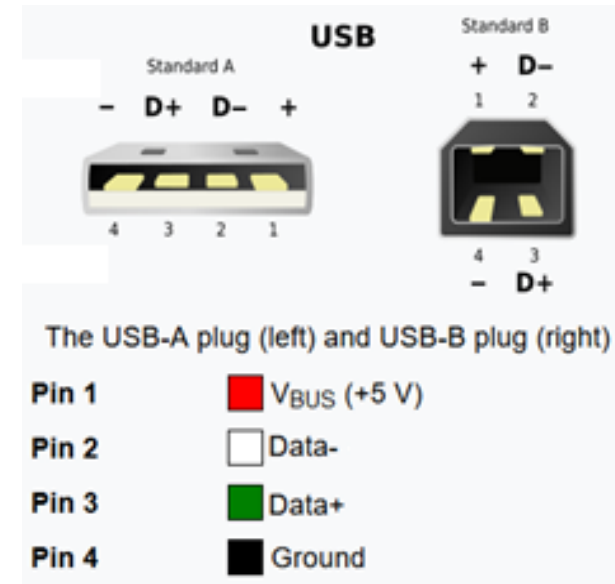
Introduction

- This chapter introduce some of the common
 - Wired and wireless communication technology used in computers.
 - USB and HDMI
 - WiFi and Bluetooth
 - User interface devices used to transfer real world data.
 - Keyboard/Mouse
 - Capacitive Touch devices
 - Camera
 - Microphone
 - Display
 - Audio Speakers
- Only a general overview of the above will be discussed, technical details will not be covered.

Universal Serial Bus (USB)



Source: <https://www.conwire.com/blog/ultimate-guide-usb-cables/>

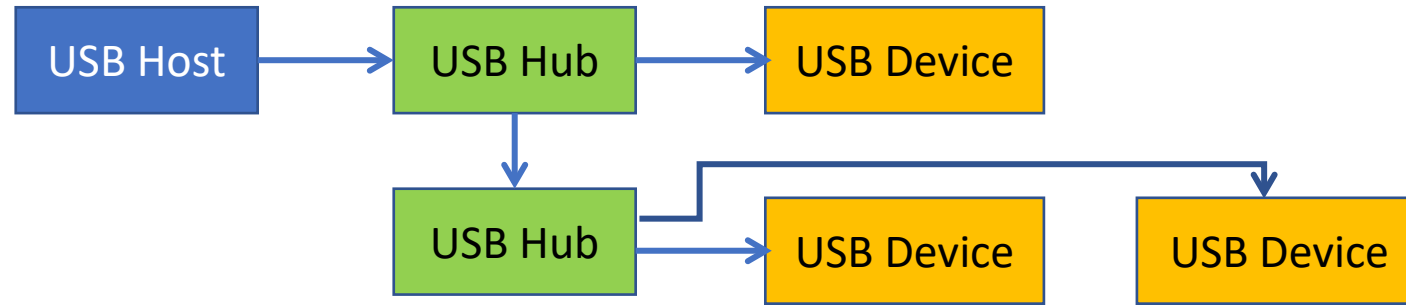


Source: <https://electronics.stackexchange.com>

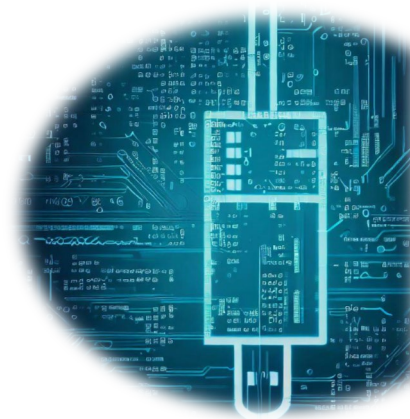
- **Serial Bus** designed to standardize connection of computer peripheral devices (Keyboard, Mouse, Printers, Disk Drives, Network Adapters, Digital Cameras etc).
- Effectively **replaced many interface buses** e.g. Serial Bus (COM Port), Parallel Port etc
- USB1.0 standard only supports a transfer rate of 12Mbps but has progressed over the years and the latest **USB3.2** standard supports up to **20Gbps** transfer.
- **4 basic pinout**: **VBUS** (+5V Power supply), **Data+/Data-** (Data pins) and **Ground** pin.



USB Topology and interface

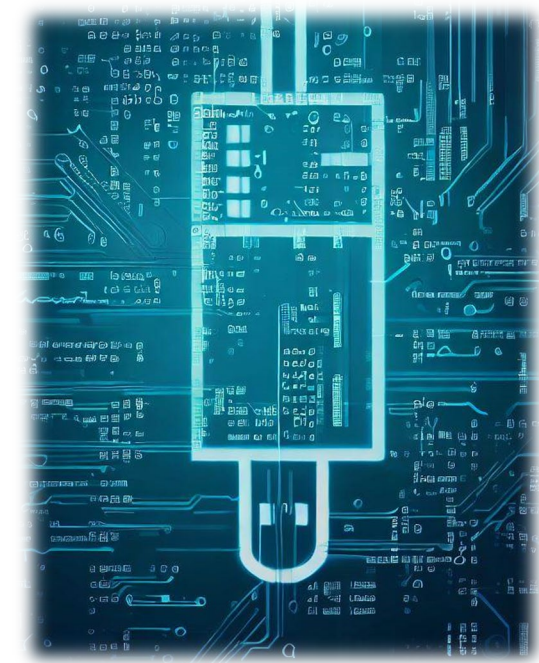


- USB used a **Tiered-Star Topology**, the center of the star is the USB host.
- All **data transactions** are **initiated by the USB Host** (typically resided on the computer). USB Peripherals (Mouse, Keyboard etc) can be connected directly to the **Host** or indirectly via the **USB Hub** devices.
- **Host will assign address to devices** connected to it to enable proper communication.
- All **data transaction** is **with respect to the USB Host**, i.e. data going into the Host is known as IN transaction and data going out of the Host is known as OUT transaction.
- **Power** to the devices can be **supplied by the Host or Hub**, known a **bus-powered**, or the device could have its **own power source (self-powered)**.



USB Enumeration and Device Class

- When the USB device is first connected to the Host, it will undergo an enumeration process where the device and the host will exchange information on their capability and requirement. E.g. a USB mouse when connected informs the Host its Vendor and Product ID, the device class it support, bandwidth required, etc.
- The host cannot communicate with the device until it is properly enumerated.
- Once all the device information is transferred to the Host, the host will check if it has the required device driver to support the USB device according to the USB device class that the device belongs to.
- There are many USB Device Classes, common ones are
 - Human Interface Device (HID) used in Mouse/Keyboard.
 - Communication Device Class (CDC) used to implement Virtual COM Port e.g. Arduino Board, MSP432 Launchpad used in the lab.
 - Mass Storage Class (MSC) used to interface to external USB HDD/SSD.
 - USB Audio Class used to stream audio to USB headset/Microphone.



High-Definition Multimedia Interface (HDMI)

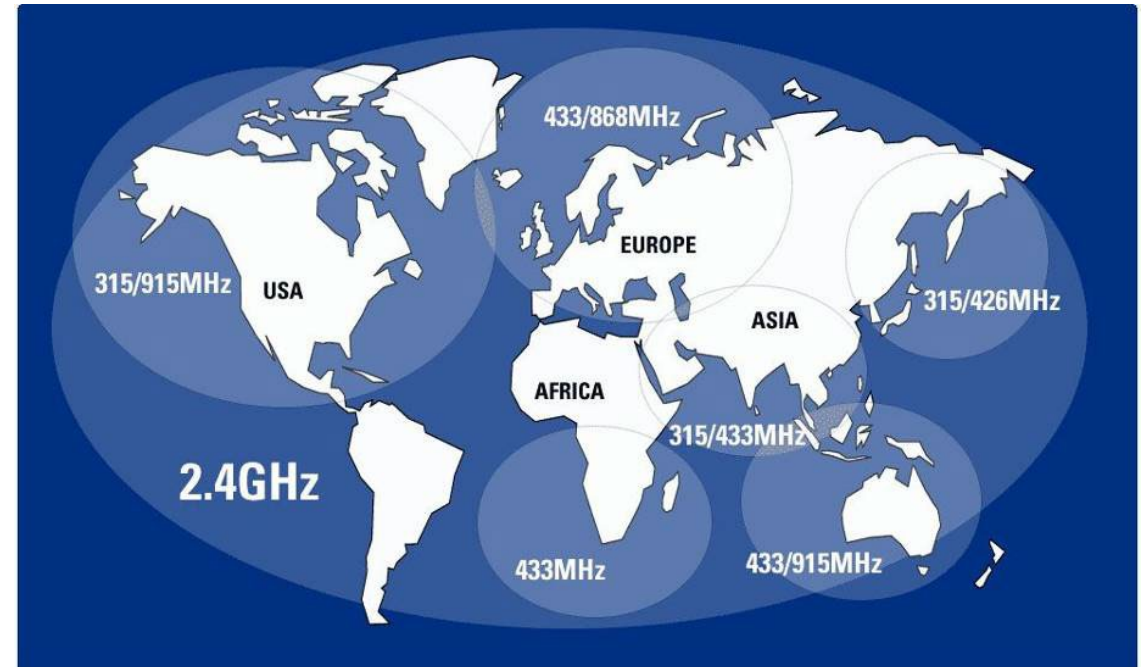
- HDMI is a **proprietary audio/video interface** for transmitting **uncompressed video data and compressed/uncompressed digital audio data** from a source device, such as a display controller in a computer, to a compatible HDMI receiver such as computer monitor, digital television, or digital audio device.
- In addition to transferring audio/video data, the **CEC (Consumer Electronics Control) capability allows HDMI devices to control each other** when necessary and allows the user to operate multiple devices with one handheld remote control device.
- Three commonly used type of HDMI connector type are shown in the figure on the right. Starting from the left: **Type D (micro)**, **Type C (Mini)** and **Type A**.
- The original HDMI v1.0/1.1 can only support a transfer rate of 3.96Gbps, allowing video format up to 1080p 60fps.
- The **latest HDMI v2.1 can achieve 42.6Gbps**, allowing **8K resolution video** to be displayed.



Source: <https://eocortex.com>

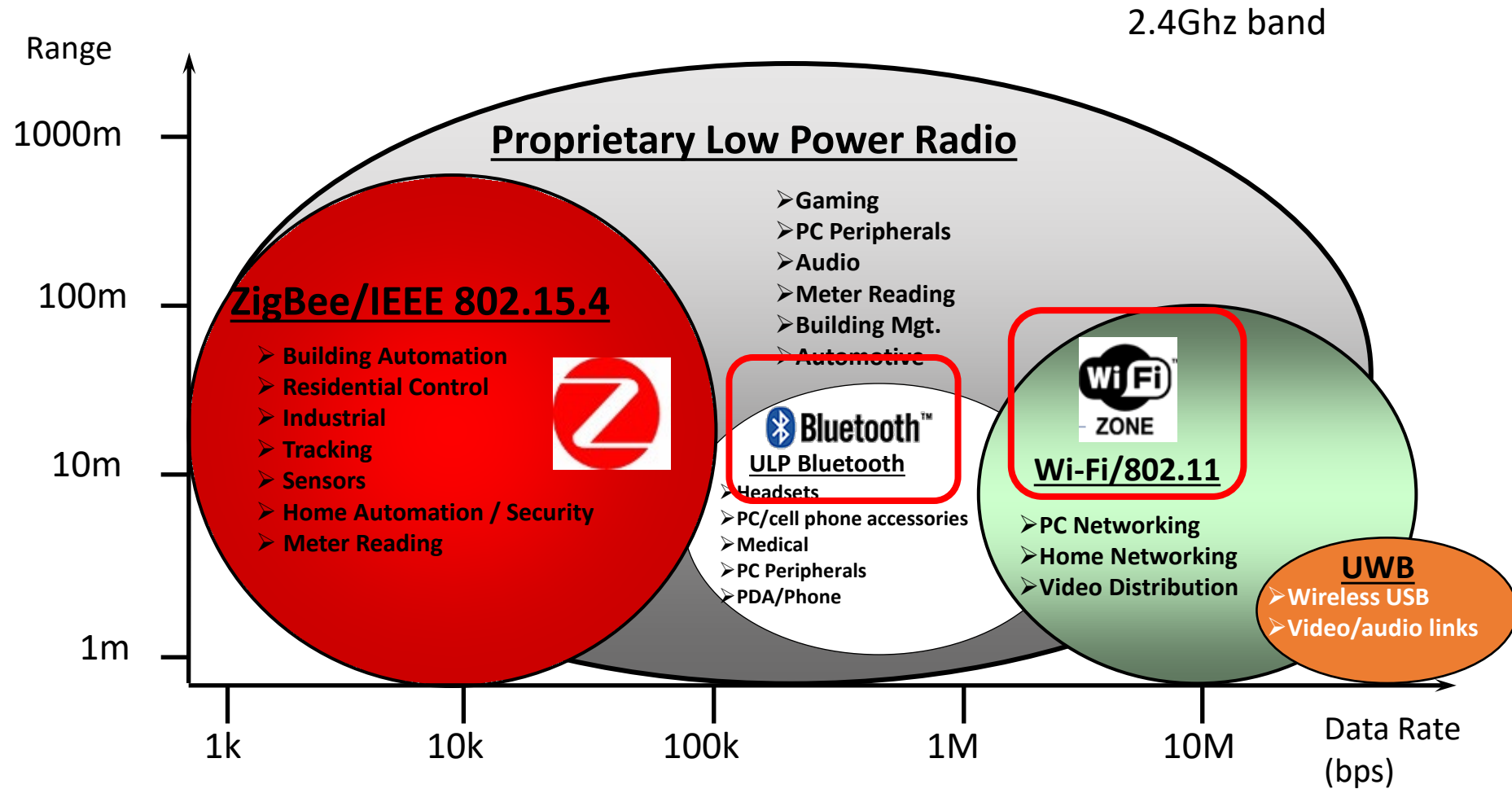
Industrial, Scientific and Medical (ISM) RF Band

- A range of “Royalty Free” Radio Frequency Bandwidth
- Some are applicable world wide while some are restricted to certain geographical regions.
- The two commonly known world wide ISM bands are 2.4Ghz and 5.8Ghz.



Source: <https://predictabledesigns.com/most-important-decision-when-creating-wireless-product/>

Wireless Standards in ISM



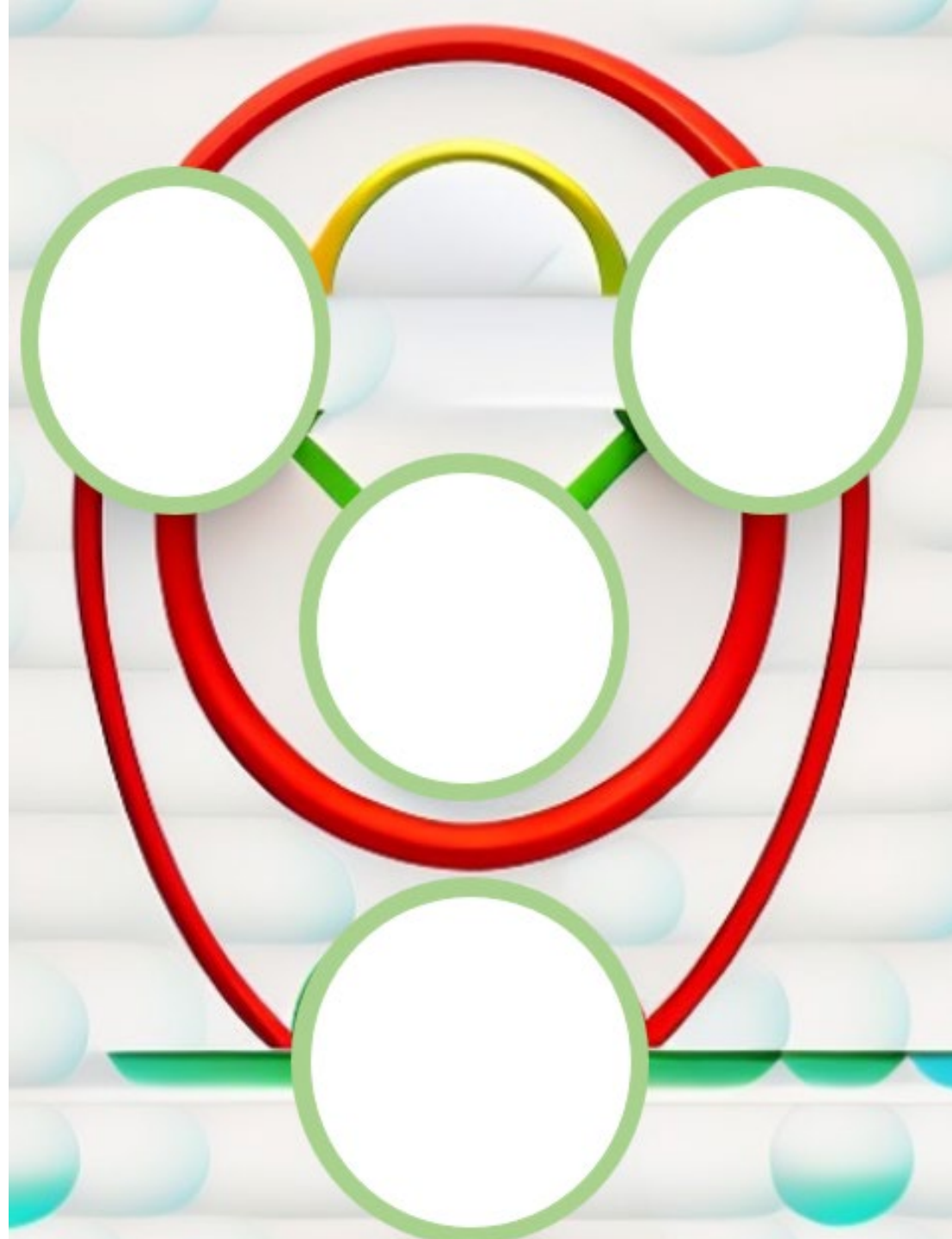
Source: Texas Instruments

Pic Source: Texas Instruments

WiFi

- A family of wireless networking technologies, based on the IEEE 802.11 family of standards, commonly known as “Wireless LAN”.
- Operates in the 2.4Ghz and 5.8Ghz RF range.
- Two common topologies: Infrastructure and Adhoc.
- Infrastructure Mode
 - Uses Star topology, at the center of the network is an Access Point or Router, connected to devices at the end.
- Adhoc Mode
 - Peer to Peer connection
- Transmission Range generally between 20m to 150m, factors affecting the range include transmission frequency, transmission power and interference.
- Transfer rate of up to ~10Gbps for the latest 802.11ax (WiFi 6).





Bluetooth



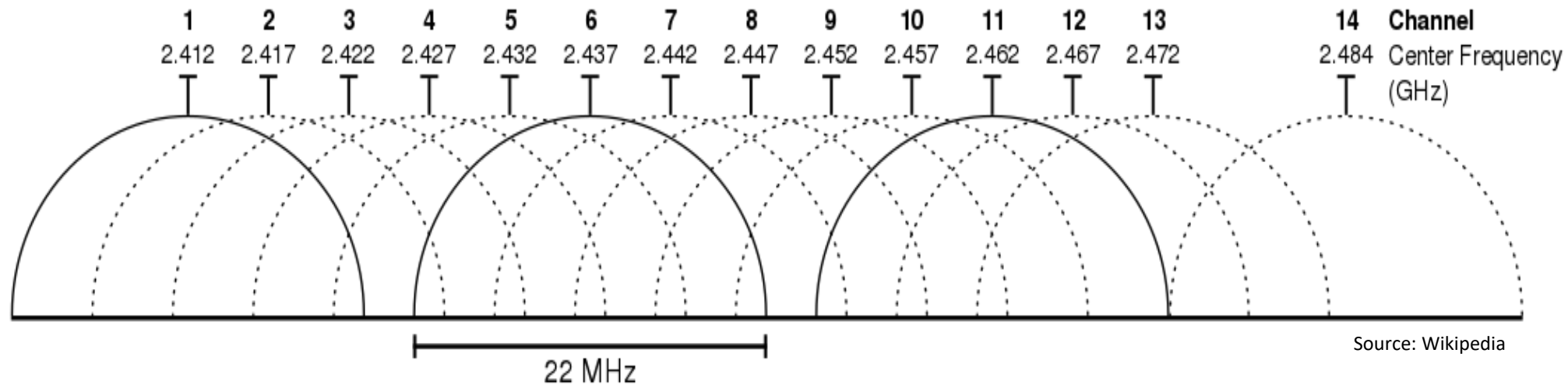
- Mainly used for low data rate wireless transmission with focus on low power consumption.
- Example: bluetooth headset, smart wearables, Bluetooth mouse etc.
- Operates in 2.4Ghz range
- Transmission range up to 100m but typically kept to 10-20m to keep power consumption low. Factors affecting transmission range is similar to that of WiFi.
- Star topology
- Transfer rate in order of Kbps and Mbps.
- The Bluetooth standard defined two different Bluetooth protocols: Bluetooth Classic (BT Classic) and Bluetooth Low Energy (BLE).
- These are based on two completely different network protocols and are not compatible. But most Bluetooth Hosts today are “Dual Mode” host so is able to connect to both BT Classic and BLE devices.



Factors affecting transmission range

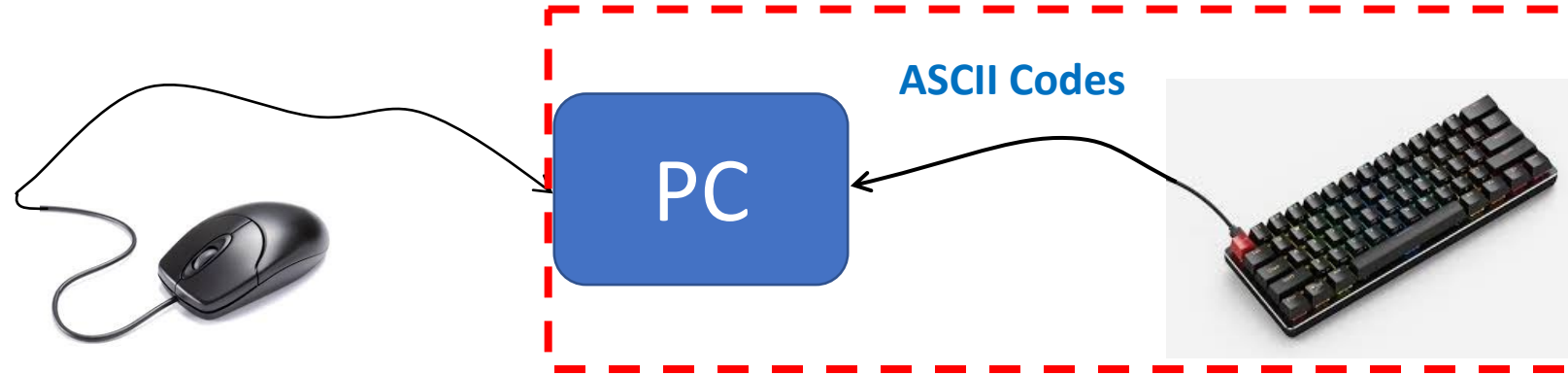
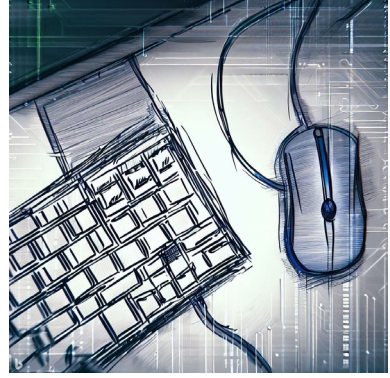
- Transmission power
 - Transmission range increase as transmission power increase.
- Transmission frequency
 - Higher frequency signal experience higher attenuation when propagating through the air or other medium.
 - All things equal, higher frequency signal has lower range than lower frequency signal.
- Interference
 - Many commonly adopted standards such as Wifi and Bluetooth works in the same 2.4Ghz ISM band. Their transmission will interfere with each other.
 - The closer the transmitter is to each other, the stronger the interference.
 - Even the micro oven in your kitchen operates in the 2.4Ghz range.

Mitigating Interference (some methods)



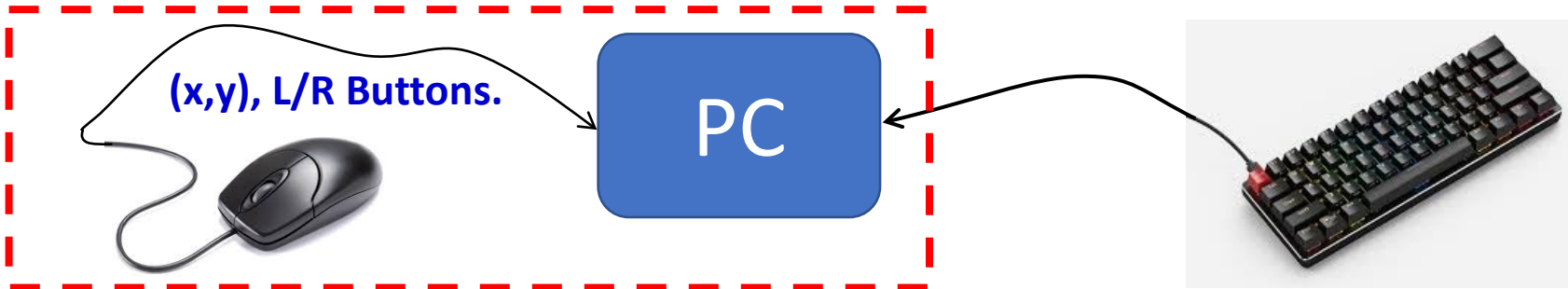
- “2.4Ghz ISM band” really consist of a [band of frequencies between 2.4 and 2.5Ghz](#). Similarly, “5.8Ghz band” also span across a certain frequency range.
- Figure above shows the Wifi standard dividing the given frequency band into sub-bands known as [channels](#).
- One common way to mitigate Wifi interference is to [select different channels](#) to use from your neighbours.
- Another common way is to use [frequency hopping](#), i.e. to constantly hop from one channel to another so that transmission will eventually succeed. Bluetooth uses frequency hopping.
- There are [other methodologies and techniques](#) employed to further mitigate interference between transmitters.

Keyboard and Mouse



- **Keyboard**
- User input device for transmitting characters (alphabets, numbers, special symbols etc).
- Has a small micro-controller on board to detect the key press and sent their corresponding **ASCII codes** to the computer.
- Connection to PC is via wired or wireless means.
- **Wired** keyboard today mainly uses **USB interface**, keyboard will be enumerated as a HID Keyboard device.
- **Wireless** keyboard mainly operate in the **2.4Ghz ISM**, technology could be Bluetooth or proprietary 2.4Ghz RF protocol.

Keyboard and Mouse



- **Mouse**
 - User **pointing device**, tracks its position by mechanical or optical means.
 - Information reported are the **(x,y) coordinates and the left/right mouse button**.
 - More information may be reported for mouse with more advanced features.
 - Information reported back to PC periodically and is known as the **scan/polling rate** of the mouse.
 - Typical mouse has a scan rate of **125Hz**, gaming mouse scan rate could be up to **1000Hz** or higher. Higher scan rate typically implies better response.
 - Another parameter is the **Dot-Per-Inch (DPI)**, which measures how fine the mouse could track the physical movement, higher DPI implies **higher sensitivity to small physical mouse movements**.
 - Connection to PC utilise similar technology as Keyboard.

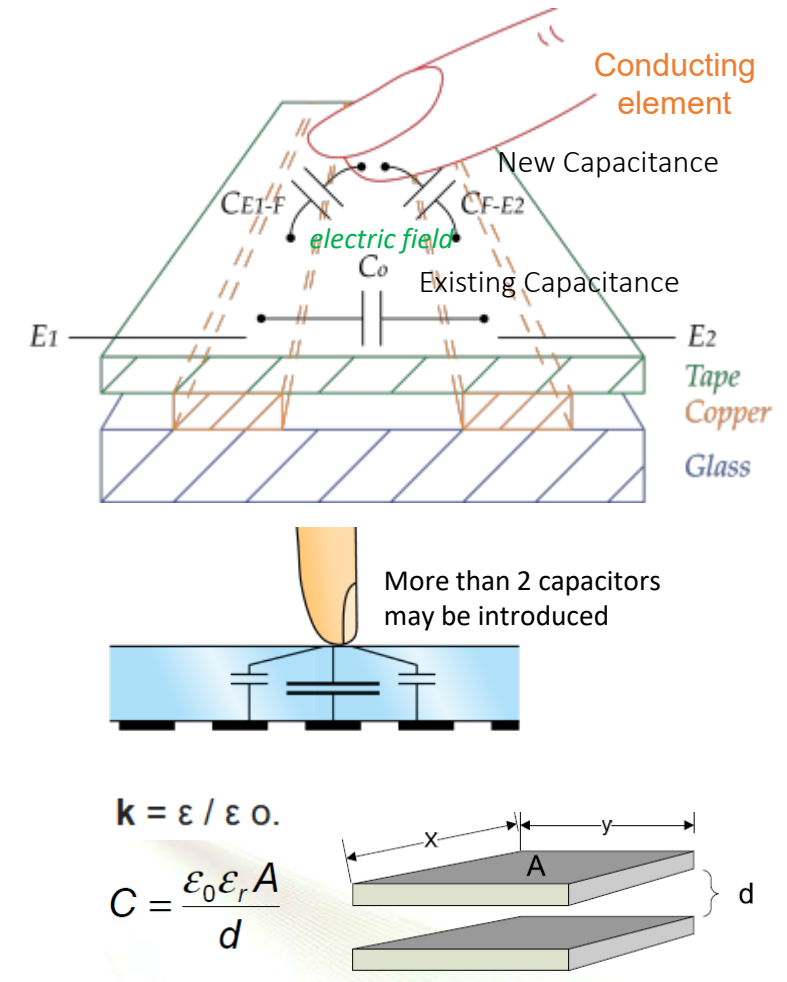
ASCII Table

MS LS	0	1	2	3	4	5	6	7
0	NUL	DLE	SP	0	@	P	'	p
1	SOH	DC1	!	1	A	Q	a	q
2	STX	DC2	"	2	B	R	b	r
3	ETX	DC3	#	3	C	S	c	s
4	EOT	DC4	\$	4	D	T	d	t
5	ENQ	NAK	%	5	E	U	e	u
6	ACK	SYN	&	6	F	V	f	v
7	BEL	ETB	'	7	G	W	g	w
8	BS	CAN	(	8	H	X	h	x
9	HT	EM	)	9	I	Y	i	y
A	LF	SUB	*	:	J	Z	j	z
B	VT	ESC	+	;	K	[	k	{
C	FF	FS	,	<	L	\	l	
D	CR	GS	-	=	M	]	m	}
E	SO	RS	.	>	N	^	n	
F	SI	US	/	?	O	_	o	DEL

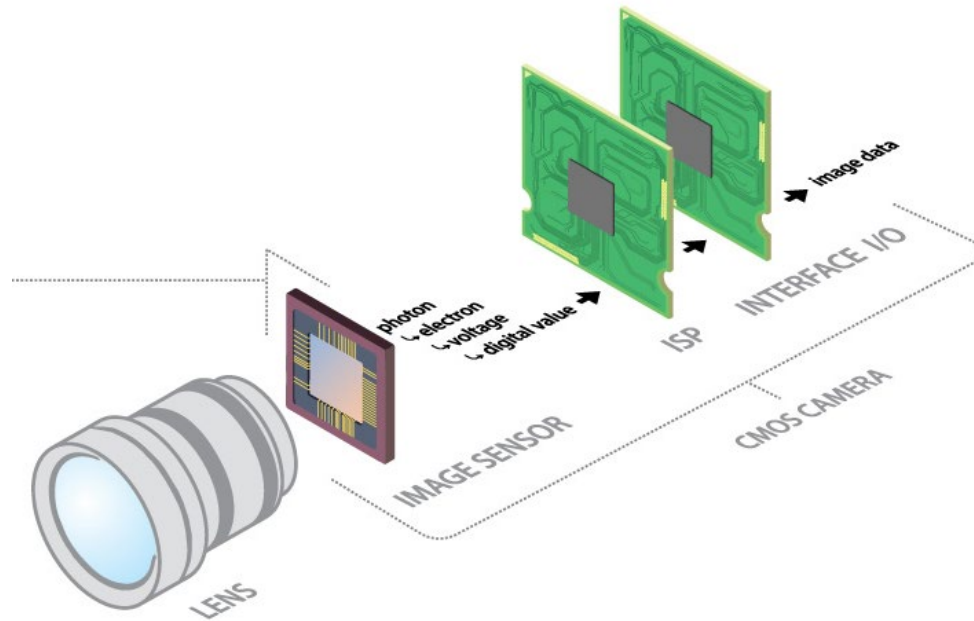
ASCII Character Set (7-Bit Code)

Capacitive Touch Interface

- A **capacitive touch pad/button** is seen as a **capacitor**.
- When a conductive element is present, e.g. finger, the **effective capacitance of the setup increases** (C_0 and combination of C_{E1-F} and C_{F-E2} : in parallel)
- Capacitance is also affected by any **dielectric** e.g. gloves, plastics, liquid between finger & pad.
- Capacitance is **directly proportional to dielectric constant (k)** and air typically has a smaller dielectric constant (~ 1), compared to all other materials (>1).
- That's why your phone touch screen, which is typically capacitive touch based, don't work as well if you have a glove on or the screen is wet.
- **Calibration** is done under assumption of human finger touch; A **processor** is used to process the information to derive the **exact position** of contact.



Camera



Source: LUCID VISION LABS



Source: Thinklucid.com, wired.com

- Typical camera on phones or PC today uses **CMOS** camera module, consisting of the sub-modules shown in the figure above.
- Sub-modules: **Lens, Image Sensor, Image Signal Processor and Interface I/O.**

Camera Sub-Modules

- Lens
 - Optical lens to focus the real world objects onto the image sensor.
- Image Sensors
 - Mostly CMOS based these days.
 - transduce photons (light information) to electrical signals (analog).
 - Analog-to-Digital conversion of the analog electrical signal to its digital equivalent for processing in the ISP.
- Image Signal Processor (ISP)
 - Processor designed to specifically handle image processing of collected image data from the sensor.
 - Processing done include Auto Focus, Auto Exposure, Auto White Balance, image format conversion, image post-processing/compression etc.
 - Common image formats are YUV (luminance and chrominance) and RGB (Red, Green, Blue).
- Interface I/O
 - Re-formatting of image data from the ISP for delivery to the host processor.



Related



Alcatel-Lucent/Bell Labs Camera
Men: Willard Boyle [left] &
George Smith [right] in 1974

CCD camera inventors

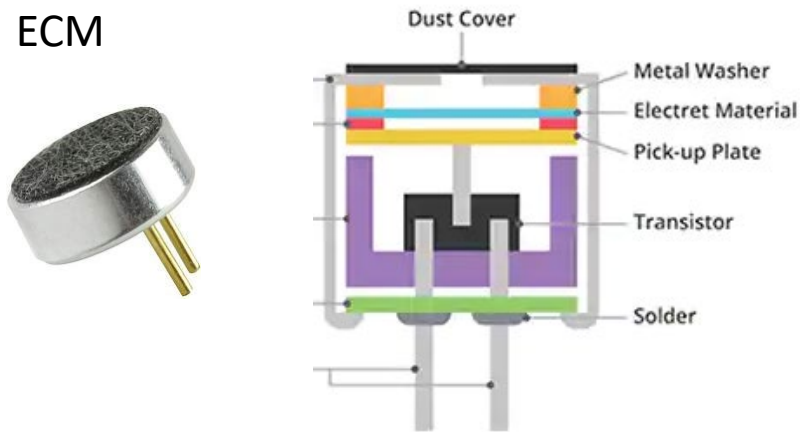
Winners of Nobel Prize in Physics (2009): George Smith and Willard Boyle.

Ubiquity of images and videos:

- ☐ Enabled by [image/video coding](#) technology
- ☐ Huge economic & social [impact](#)

Microphones

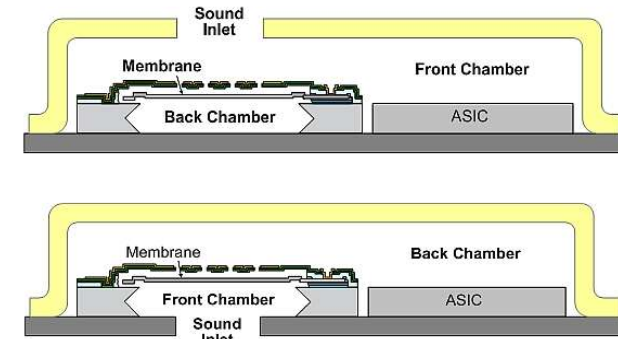
ECM



MEMS Microphone



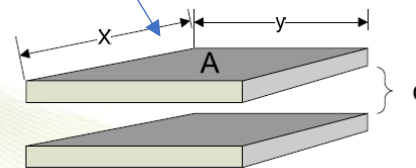
Source: EDN, DigiKey



- Two of the most popular types of microphones are **micro-electro-mechanical system (MEMS) microphones** and **electret condenser microphones (ECM)**.
- Both MEMS and ECM microphones use the **variation in capacitance** when the **diaphragm is displaced by the sound pressure** to transduce sound wave to electrical signals.

$$k = \epsilon / \epsilon_0$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

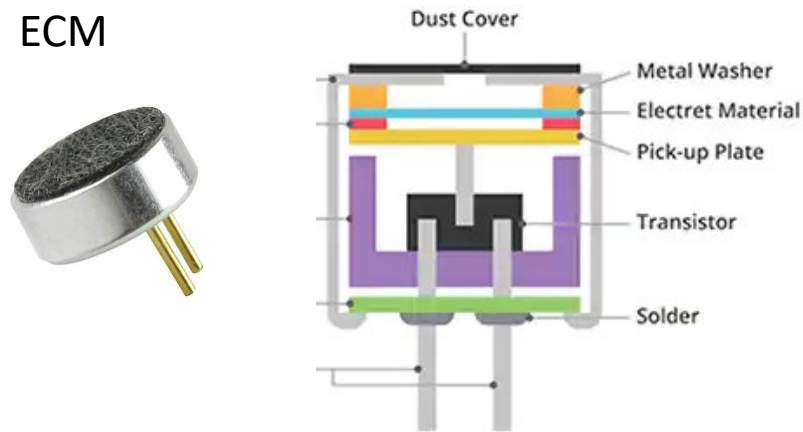


d change $\rightarrow$ C change $\rightarrow$ V change

- The transduced signal is analog in nature but an ADC could be added to enable a digital output.

Microphones

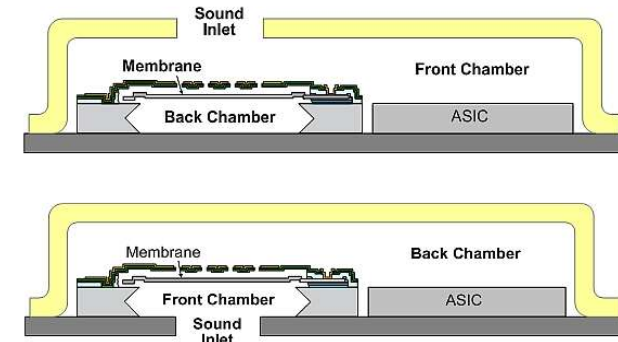
ECM



MEMS Microphone



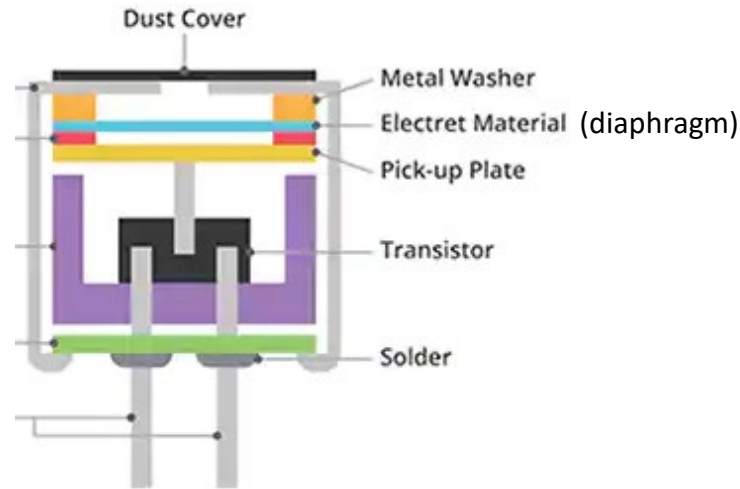
Source: EDN, DigiKey



- Difference is the in ECM, the electrical charges needed to measure the change in capacitance is provided by the charges stored on the electret.
- In MEMS Microphone case, the electrical charges needed is provided by a charge pump instead.



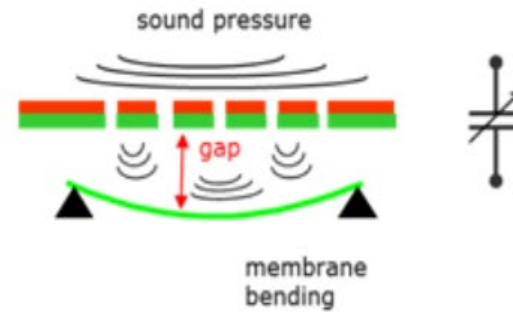
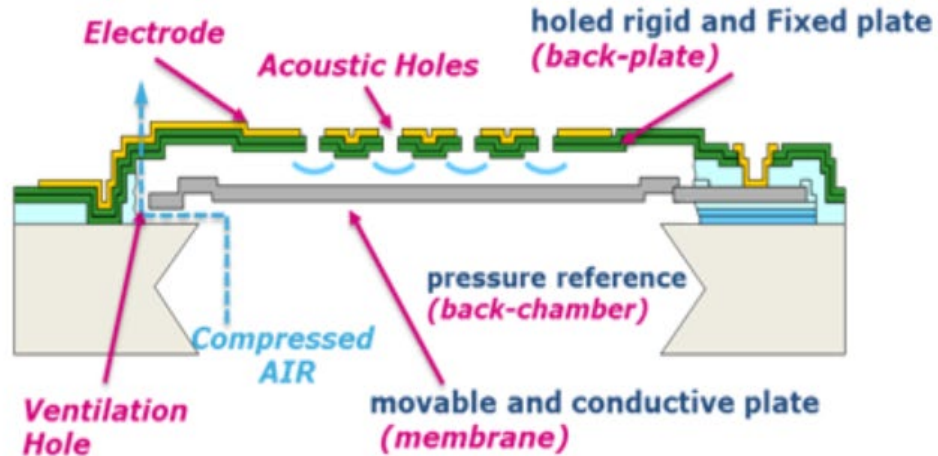
ECM Microphone



- In an ECM, the electret diaphragm is a material with a fixed surface charge that's placed near a conductive plate
- A capacitor is created with the air gap forming the dielectric.
- Sound pressure waves moving the electret diaphragm cause the value of the capacitance to change, causing voltage across the capacitor to vary, $V = Q/C$ (Q = a fixed charge).



MEMS Microphone

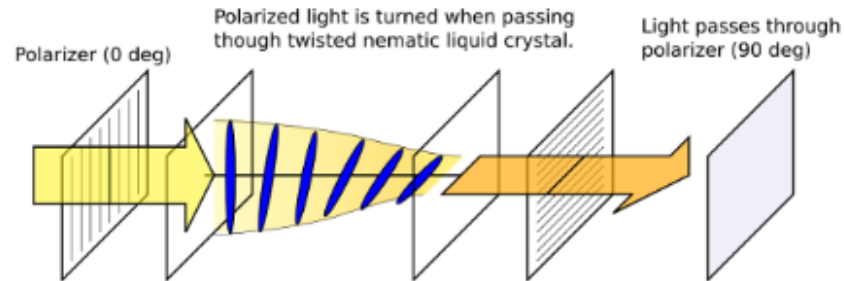


- MEMS microphone basically is an acoustic transducer.
- Transduction principle is the coupled capacity change between a fixed plate (back-plate) and a movable plate (membrane)
- The capacitive change is caused by the sound, passing through the acoustic holes, that moves the membrane modulating the air gap comprised between the two conductive plates



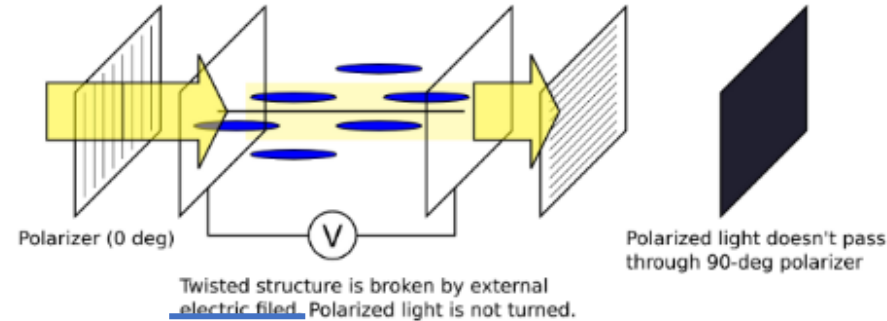
Liquid Crystal Display (LCD) Basics

Polarised light twisted (Light Passes)



Source: awawa.hariko.com

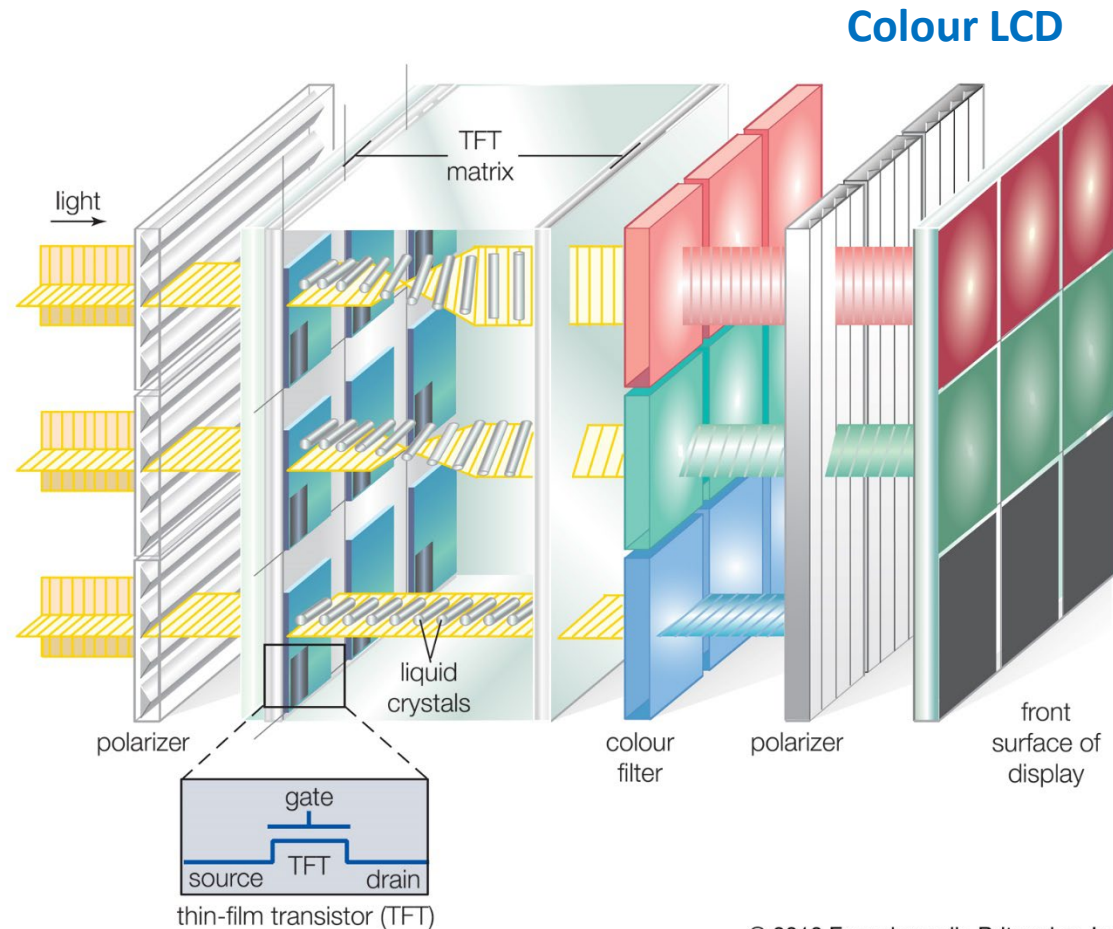
Polarised light not twisted (Light Blocked)



- **Liquid crystal** is an organic substance that has both a liquid form and a crystal molecular structure. The rod-shaped molecules are **able to keep their order in a particular direction**.
- An **electric field** can be used to **control the molecules orientation**, according to pixel value (brightness) of an image.
- Depending on their orientation, these molecules are able to **twist the light passing through them**.
- The **amount of light** that is able to pass through the polariser **depends on its orientation with respect to the polariser**.
- This results in **light passing, partial passing or being blocked** from user point of view.



Colour LCD

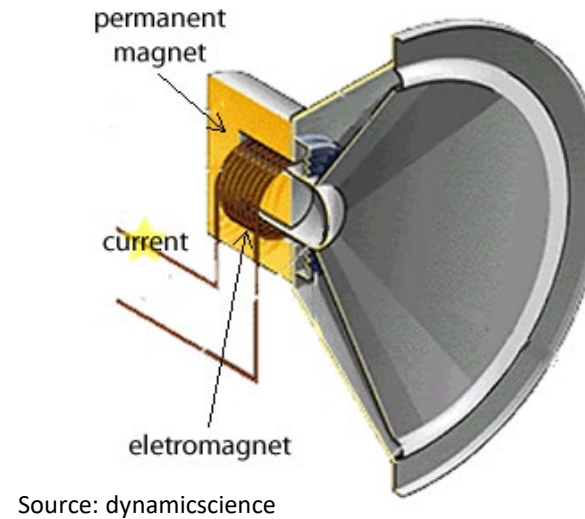
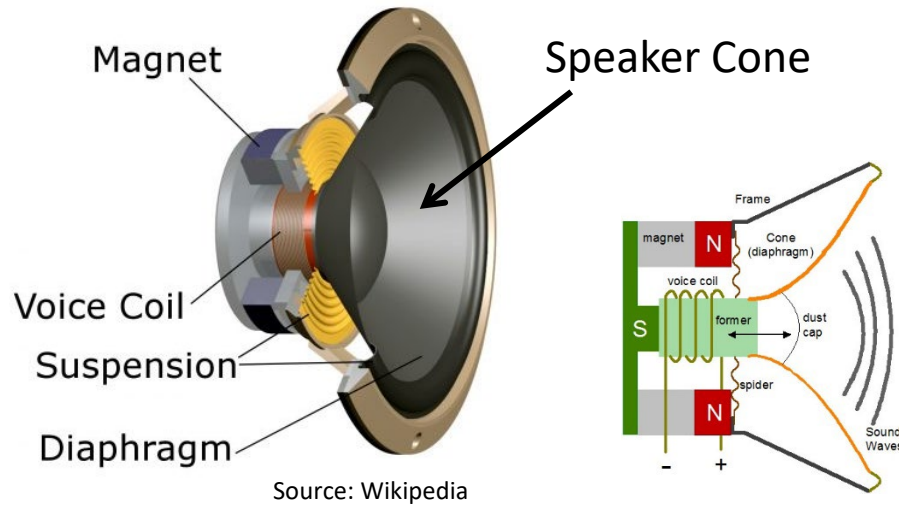


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- Process for colour LCD is similar to that discussed in previous slide.
- Each pixel is **associated with three colour filters** to project the RGB colours.
- A **varying orientation** of molecules in the liquid crystal suspension: **varies the amount of light** allowed to pass through to the colour filter, thereby changing the colour picture on the display screen.



Audio Speaker



- The device **transduces electrical energy to sound energy**.
- The **speaker cone** vibrates, pushing and pulling the air to create sound waves.
- The conversion **from electrical to mechanical energy** occurs through an electromagnetic coil and magnet combination attached to the cone. This coil **moves the speaker cone back and forth** as its electromagnetic field changes with the electrical current passing through it, **converting the mechanical energy to sound energy**.



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